
Data stream
$p_t \neq p_{t+1}$
t
$(\mathbf{x}_t, y_t) \sim p_t$
Adaptation strategies
1 Classical streaming
$\hat{y}_t$
θ_{t+1}
θ_t
$(\mathbf{x}_t, y_t)$
$\theta_{t+1} \leftarrow U(\theta_t, (\mathbf{x}_t, y_t))$
2 Streaming AutoML
arg max
prep
model
prep
model
prep
model
$\Pi_{t+1} \leftarrow \arg\max_{\Pi \in \mathcal{P}} \text{score}(\Pi)$
3 In-context models
$\mathbf{x}_t$
θ
M_t
$\hat{p}(y \mid \mathbf{x}_t, M_t)$, θ fixed
One protocol
predict $\hat{y}_t = f(\mathbf{x}_t)$
observe y_t
t
$|M_t| \leq B$, q queries, delay d
Accuracy
Cost